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1 **(a)** A liquid L fills a container of very large uniform cross-sectional area to a certain depth. Another liquid M is now added to the container. The two liquids do not mix as shown in Fig. 1.1. The total depth of the liquids is 0.17 m.

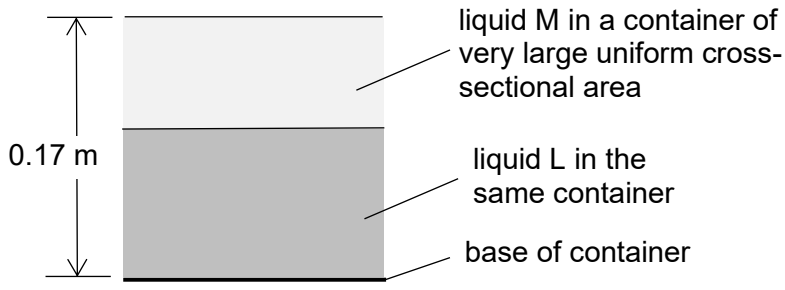

Fig. 1.1 (not to scale)

Fig. 1.2 shows how the pressure p inside the liquids varies with height x above the base of the container.

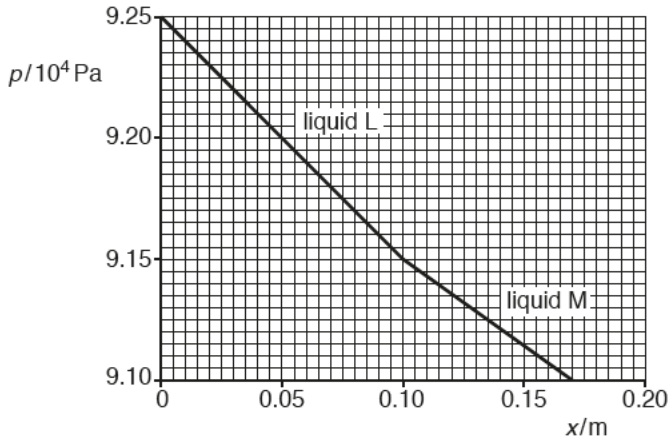

Fig. 1.2

Use Fig. 1.2 to

(i) state the value of the atmospheric pressure,

atmospheric pressure = ... Pa [1]

(ii) determine the density of liquid M.

density = ... kg m^{-3} [2]

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(b) Above the liquids, a spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially held so that the spring has zero extension as shown in Fig. 1.3.

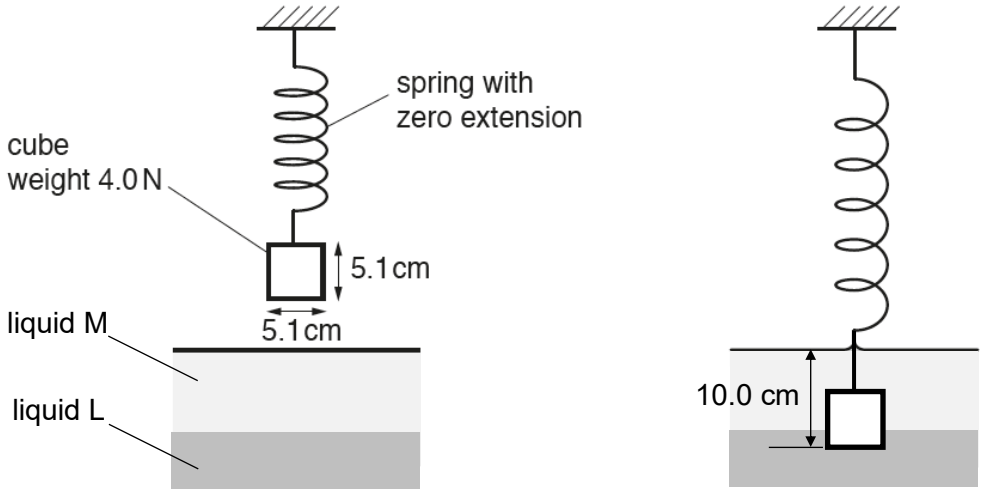

Fig. 1.3 (not to scale)

The cube has weight 4.0 N and sides of length 5.1 cm. The cube is released and sinks into the liquids as the spring extends. The cube reaches equilibrium with its base at a depth of 10.0 cm below the top surface of the liquid M, as shown in Fig. 1.3.

(i) Determine the upthrust acting on the cube.

upthrust = ... N [3]

(ii) Calculate the magnitude of the force exerted on the spring by the cube when it is in equilibrium in the liquids.

force = ... N [1]

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(c) Suggest how to check that the elastic limit of the spring is not exceeded.

……

…… [1]

(d) Two identical balls are placed in a smooth glass container as shown in Fig. 1.4. Each ball has a mass of 170 g. Their centres and point A lies on a straight line as shown by the dotted line.

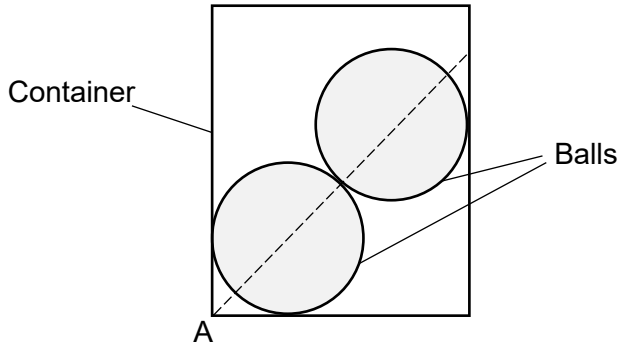

Fig. 1.4

(i) Determine the magnitude of the horizontal force by the container on the upper ball.

horizontal force = …… N [2]

(ii) Hence, or otherwise, determine the magnitude of the force exerted by the lower ball on the upper ball.

force = …… N [2]